

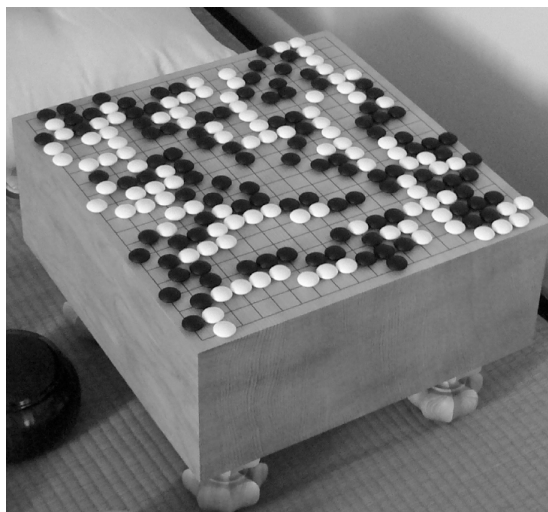


Figure 13.1: The game of Go

Picture *from*
[https://en.wikipedia.org/wiki/Go_\(game\)#media/File:FloorGoban.JPG](https://en.wikipedia.org/wiki/Go_(game)#media/File:FloorGoban.JPG)

After its opponent moved, AlphaZero would pause, select moves to consider, and then play out thousands of simulations of how the entire game might go given those selected moves. After running a set of simulations, AlphaZero might see that it won thirty-five out of the forty imagined games when it made move A, thirty-nine of the forty imagined games when it made move B, and so on for many other possible next moves. AlphaZero could then pick the move where it had won the highest ratio of imagined games.

Doing this, of course, comes with the search problem; even armed with Google's supercomputers, it would take well over a million years to simulate every possible future move from an arbitrary board position in Go. And yet AlphaZero ran these simulations within half a second. How? It didn't simulate the trillions of possible futures; it simulated only *a thousand* futures. In other words, it *prioritized*.

There are many algorithms for deciding how to prioritize which branches to search through in a large tree of possibilities. Google Maps uses such an algorithm when it searches for the optimal route from point A to point B. But the search strategy used by AlphaZero was different and offered unique insight into how real brains might work.

We already discussed how in temporal difference learning an actor learns to predict the best next move based on a hunch about the board position, doing so without any planning. AlphaZero simply expanded on this architecture. Instead of picking the single move its actor believed was the best next move, it picked multiple top moves that its actor believed were the best. Instead of just assuming its actor was correct (which it would not always be), AlphaZero used search to *verify* the actor's hunches. AlphaZero was effectively saying to the actor, "Okay, if you think move A is the best move, let's see how the game would play out if we did move A." And AlphaZero then also explored other hunches of the actor, considering the second and third best moves the actor was suggesting (saying to the actor, "Okay, but if you didn't take move A, what would your next best hunch be? Maybe move B will turn out even better than you think").

What is elegant about this is that AlphaZero was, in some sense, just a clever elaboration on Sutton's temporal difference learning, not a reinvention of it. It used search not to logically consider all future possibilities (something that is impossible in most situations) but to simply verify and expand on the hunches that an actor-critic system was already producing. We will see that this approach, in principle, may have parallels to how mammals navigate the search problem.

While Go is one of the most complex board games, it is still far simpler than the task of simulating futures when moving around in the real world. First, the actions in Go are *discrete* (from a given board position, there are only about two hundred possible subsequent next moves), whereas in the real world actions are *continuous* (there are an infinite number of possible body and navigational paths). Second, the information about the world in Go is deterministic and complete, whereas in the real world it is noisy and incomplete. And third, the rewards in Go are simple (you either win or lose the game), but in the real world, animals have competing needs that change over time. And so, while AlphaZero was a huge leap forward, AI systems are still far from performing planning in environments with a continuous space of actions, incomplete information about the world, and complex rewards.

However, the most critical advantage of planning in mammalian brains over modern AI systems like AlphaZero is not their ability to plan with continuous action spaces, incomplete information, or complex rewards, but instead simply the mammalian brain's ability to flexibly change its

approach to planning depending on the situation. AlphaZero—which applied only to board games—employed the same search strategy with every move. In the real world, however, different situations call for different strategies. The brilliance of simulation in mammal brains is unlikely to be some special yet-to-be-discovered search algorithm; it is more likely to be the flexibility with which mammal brains employ different strategies. Sometimes we pause to simulate our options, but sometimes we don't simulate things at all and just act instinctually (somehow brains intelligently decide when to do each). Sometimes we pause to consider possible futures, but other times we pause to simulate some past event or alternative past choices (somehow brains select when to do each). Sometimes we imagine rich details in our plans, playing out each individual detailed subtask, and sometimes we render just the general idea of the plan (somehow brains intelligently select the right granularity of our simulation). How do our brains do this?

Prefrontal Cortex and Controlling the Inner Simulation

In the 1980s, a neuroscientist named Antonio Damasio visited one of his patients—referred to as “L”—who had suffered a stroke. L lay in bed with her eyes open and a blank expression on her face. She was motionless and speechless, but she wasn't paralyzed. She would, at times, lift up the blanket to cover herself with perfectly fine motor dexterity; she would look over at a moving object, and she could clearly recognize when someone spoke her name. But she did and said nothing. When looking into her eyes, people said it seemed that “she was there but not there.”

Stroke victims with damage to the visual, somatosensory, or auditory neocortex suffer from impairments to perception (such as blindness or deafness). But L showed none of these symptoms; her stroke occurred in a specific region of her *prefrontal* neocortex. L had developed akinetic mutism—a tragic and bizarre condition caused by damage to certain regions of the prefrontal cortex in which people are able to move and understand things just fine but they don't move, speak, or care about anything at all.

After six months, as with many stroke patients, L began to recover as other areas of her neocortex remapped themselves to compensate for the damaged area. As L slowly began to speak again, Damasio asked her about her experience over the prior six months. Although L had little memory of